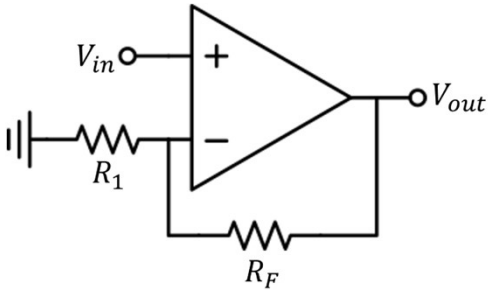
Hes·so VALAIS WALLIS
Haute Ecole d'Ingénierie
Hochschule für Ingenieurwissenschaften

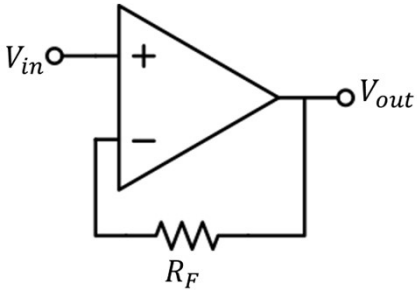
Circuits Amplificateur Opérationnel
Operationsverstärkerschaltungen

Amplificateur Non-Inverseur
Nichtinvertierender Verstärker
(Non-Inverting Amplifier)



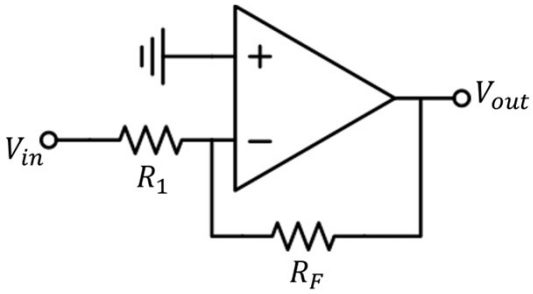
$$V_{out} = \frac{R_1 + R_F}{R_1} V_{in}$$

Amplificateur Suiveur
Spannungsfolger
(Voltage Follower)



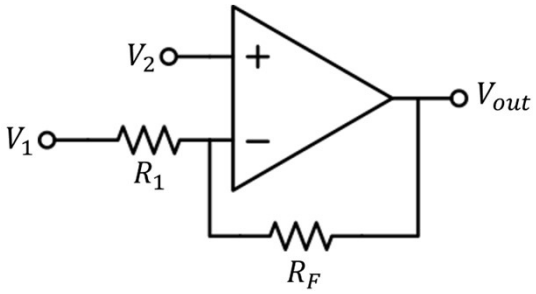
$$V_{out} = V_{in}$$

Amplificateur Inverseur
Invertierender Verstärker
(Inverting Amplifier)



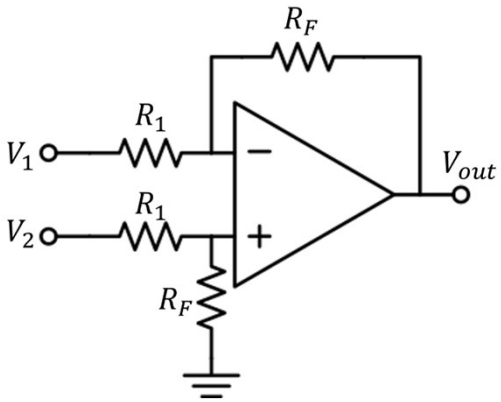
$$V_{out} = -V_{in} \frac{R_F}{R_1}$$

Cas général
Allgemeiner Fall



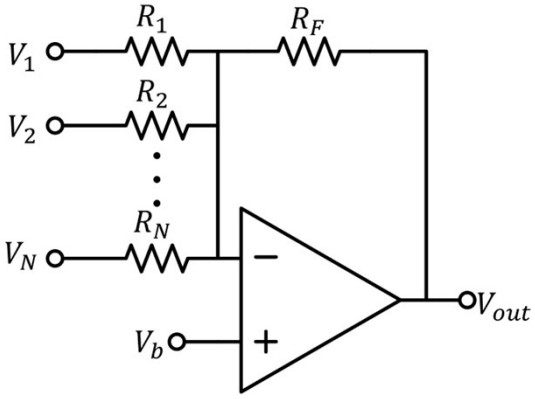
$$V_{out} = V_2 \frac{R_1 + R_F}{R_1} - V_1 \frac{R_F}{R_1}$$

Amplificateur différentiel
Differenzverstärker
(Differential Amplifier)



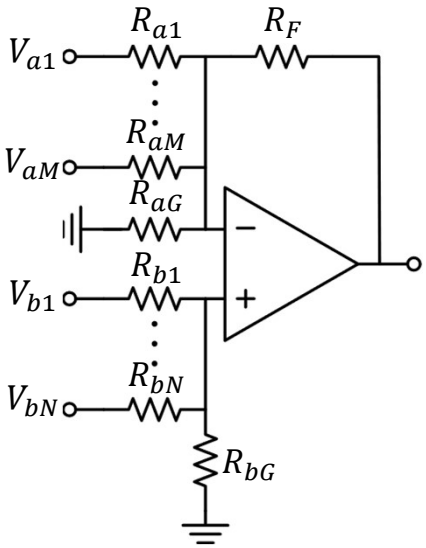
$$V_{out} = (V_2 - V_1)\frac{R_F}{R_1}$$

Inverseur sommateur
Invertierender Addierer
(Inverting adder)



$$V_{out} = V_b\left(\frac{R_F}{R_1 \parallel \cdots \parallel R_N} + 1\right) - \left(\frac{R_F}{R_1}V_1 + \cdots + \frac{R_F}{R_N}V_N\right)$$

Amplificateur sommateur-soustracteur
Addierer-Subtrahierer
(Adding-Subtracting Amplifier)



$$G_{bi} = \frac{R_F}{R_{bi}} \qquad G_{ai} = \frac{R_F}{R_{ai}}$$

$$V_{out} = \sum_{i=1}^{N} G_{bi}V_{bi} - \sum_{i=1}^{M} G_{ai}V_{ai}$$

si/wenn

$R_{aG} = \infty$

$$R_{bG} = R_F \frac{1}{1 + \sum_{i=1}^{M} G_{ai} - \sum_{i=1}^{N} G_{bi}}$$

ou/oder

$R_{bG} = \infty$

$$R_{aG} = R_F \frac{1}{\sum_{i=1}^{N} G_{bi} - \sum_{i=1}^{M} G_{ai} - 1}$$